

SEDALIA AGCM, AB, Canada

WMO: 713990

Lat: 51.590N

Lon: 110.755W

Elev: 755

StdP: 92.58

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.7	-28.6	-35.3	0.1	-31.7	-32.0	0.2	-28.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	30.8	16.3	28.7	16.0	26.7	15.5	18.5	25.7	17.5	25.1	16.6	24.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.6	21.4	14.9	11.6	20.1	13.8	10.8	18.9	55.5	25.4	52.1	25.5	49.4	24.1	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-35.7	34.4	1.9	2.4	-37.1	36.1	-38.2	37.5	-39.3	38.8	-40.7	40.5	(4)
(5)			WB	-35.8	20.8	1.9	1.2	-37.2	21.7	-38.3	22.4	-39.4	23.1	-40.7	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-10.8	-12.1	-3.9	3.6	10.8	14.9	18.0	17.2	12.1	4.3	-4.4	-11.9	(6)	
	DBStd	12.54	8.23	7.86	7.88	5.42	4.30	3.37	2.94	3.35	4.77	5.15	7.29	8.21	(7)	
	HDD10.0	3269	644	618	432	197	44	3	0	1	31	188	431	679	(8)	
	HDD18.3	5584	903	852	691	441	236	112	43	62	194	434	681	937	(9)	
	CDD10.0	801	0	0	0	7	68	151	247	223	93	11	1	0	(10)	
	CDD18.3	74	0	0	0	0	1	9	32	27	6	0	0	0	(11)	
	CDH23.3	1442	0	0	0	5	78	173	518	477	186	5	0	0	(12)	
	CDH26.7	438	0	0	0	0	10	44	166	154	64	0	0	0	(13)	
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)	
(15) Precipitation	PrecAvg	330	14	8	13	21	37	73	53	43	25	14	13	11	(15)	
	PrecMax	471	47	18	36	61	80	150	128	84	83	33	44	18	(16)	
	PrecMin	172	2	1	3	3	3	17	20	5	1	4	0	2	(17)	
	PrecStd	75	10	5	7	15	23	32	24	21	21	7	11	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.2	16.4	23.5	28.0	31.3	33.1	32.7	31.8	23.3	15.8	6.3	(19)	
		MCWB	3.2	3.1	8.1	10.7	13.6	16.1	17.4	16.7	15.5	12.5	7.5	2.8	(20)	
	2%	DB	4.0	2.9	12.6	19.8	25.5	27.4	30.8	30.3	28.4	19.1	11.0	3.9	(21)	
		MCWB	1.8	0.7	6.0	9.2	12.6	15.2	17.4	16.3	14.7	10.5	5.4	1.7	(22)	
	5%	DB	2.5	1.1	9.4	16.5	23.3	25.1	28.4	28.2	25.0	16.0	7.3	1.3	(23)	
		MCWB	0.7	-0.4	4.4	7.2	11.9	14.8	16.9	15.9	13.5	8.7	3.5	-0.6	(24)	
	10%	DB	0.6	-0.8	6.0	13.5	20.6	23.1	26.2	26.0	22.0	13.1	4.5	-1.4	(25)	
		MCWB	-0.9	-2.0	2.5	5.9	10.5	14.0	16.4	15.5	12.4	7.3	1.8	-2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	3.3	8.4	10.9	14.7	18.6	20.9	19.1	16.5	13.6	8.6	3.4	(27)	
		MCDB	5.8	5.8	15.2	21.8	25.3	25.4	29.4	26.3	27.3	22.3	15.4	5.5	(28)	
	2%	WB	1.8	1.0	6.4	9.3	13.5	17.0	19.0	18.0	15.2	11.2	5.9	1.5	(29)	
		MCDB	3.7	2.6	11.7	18.8	23.6	24.4	25.7	25.9	26.3	16.9	10.2	3.8	(30)	
	5%	WB	0.7	-0.3	4.6	7.7	12.4	15.8	17.9	17.1	14.0	9.2	3.8	-0.4	(31)	
		MCDB	2.4	1.2	9.1	15.5	21.5	23.0	25.2	25.3	23.3	14.9	6.8	1.3	(32)	
	10%	WB	-0.9	-2.0	2.6	6.3	11.3	14.7	17.1	16.1	12.8	7.7	2.0	-2.6	(33)	
		MCDB	0.6	-0.9	5.6	12.7	19.2	21.4	24.2	24.1	21.2	12.9	4.3	-1.4	(34)	
(35) Mean Daily Temperature Range	MDBR		10.3	10.7	11.5	13.0	14.2	13.0	14.2	14.3	14.6	12.9	10.3	9.6	(35)	
	5% DB	MCDBR	10.5	11.6	15.7	19.2	18.8	16.7	18.3	18.9	20.2	17.6	14.4	11.9	(36)	
		MCWBR	8.6	9.8	9.5	9.6	8.5	7.1	7.0	7.1	8.6	9.7	9.5	9.7	(37)	
	5% WB	MCDBR	10.4	11.3	15.0	17.8	16.6	14.5	15.2	15.9	19.0	16.7	13.5	11.7	(38)	
		MCWBR	8.8	9.7	9.4	9.3	7.9	6.8	7.0	6.8	8.4	9.4	9.3	9.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.310	0.294	0.346	0.373	0.372	0.365	0.381	0.328	0.287	0.262	0.256	(40)	
	taud		2.338	2.224	2.437	2.309	2.275	2.333	2.388	2.307	2.442	2.530	2.470	2.330	(41)	
	Ebn at Noon		764	802	913	892	879	882	845	862	844	775	708		(42)	
	Edh at Noon		61	91	90	115	126	120	112	115	89	67	53	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.16	2.23	3.60	4.84	5.84	6.05	6.49	5.28	3.79	2.29	1.21	0.88	(44)	
	RadStd		0.11	0.19	0.27	0.40	0.39	0.37	0.28	0.39	0.35	0.29	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page